ICTP-SAIFR EGSnrc Monte Carlo School for Medical Physics

IFT-UNESP, São Paulo, Brazil

March 30 – April 3, 2026

LECTURE 05 · EGSnrc course

Photon physics

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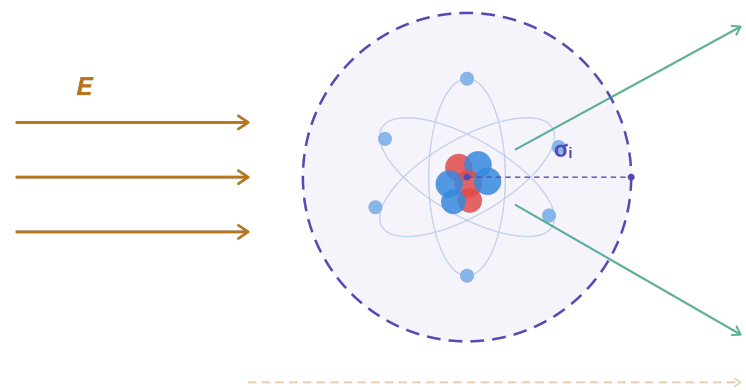
Interaction cross section $\sigma_i(E)$

Microscopic quantity related to the strength of an interaction type i.

- Measure of how strongly an incoming particle of energy E interacts with a target through a specific process i.
- Probabilistic *"effective target area"* presented by an atom to an incoming particle of energy E with respect to interactions of type i
- Emerges from *quantum scattering amplitudes*, **not** the physical size of the atom.
- Typical units:
  - cm^2 per atom
  - barns per atom
    (1 barn = 10^{-24} cm^2)



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Interaction probability

- **Total cross section** $\sigma(E)$: given by the sum of the individual interactions:

$$\sigma(E) = \sum_i \sigma_i(E)$$

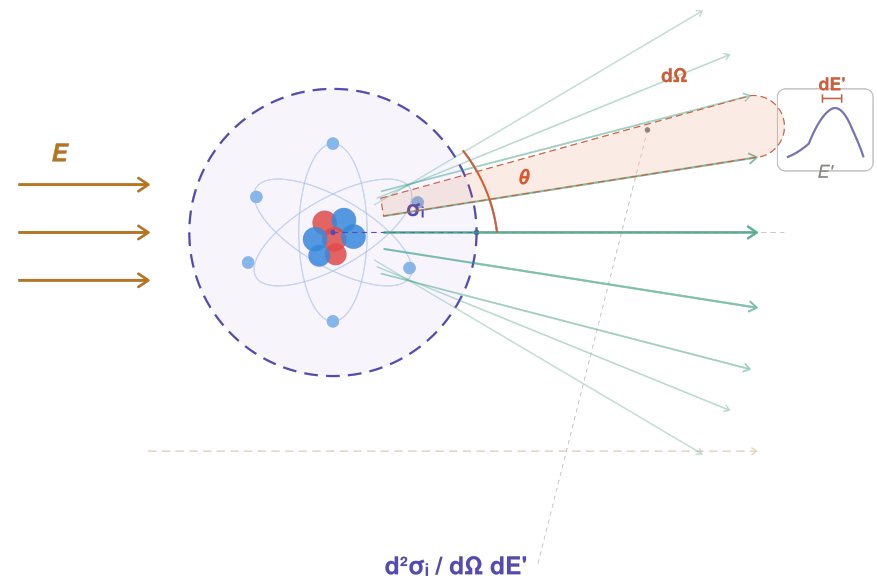
- **Interaction probabilities** (branching ratios) $w_i(E)$: probability that the interaction is of type i when a photon of energy E interacts with the surrounding medium, e.g.,

$$w_i(E) = \frac{\sigma_i(E)}{\sigma(E)}$$

Useful for sampling interaction type during simulation of radiation transport.

Double differential cross section $\frac{\mathrm{d}^2\sigma_i(E,E',\Omega)}{\mathrm{d}\Omega\mathrm{d}E'}$

- Cross section per unit energy and per unit solid angle for a given interaction outcome.
- *Everything else is built from this.*
- Not a probability density by itself, only when normalized by σ_i.



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The **most fundamental** quantity. Obtained from theoretical models and experimental measurements.

Energy and direction after an interaction

For example, the double differential cross section $\frac{\mathrm{d}^2\sigma_i(E,E',\Omega)}{\mathrm{d}\Omega\mathrm{d}E'}$ integrated over Ω and divided by the integral cross section σ_i

$$\frac{1}{\sigma_i(E)}\frac{\mathrm{d}\sigma_i(E,E')}{\mathrm{d}E'}\mathrm{d}E'$$

is the probability that a photon of energy E will scatter into energy interval $\mathrm{d}E'$ around E' after interaction type i.

Equivalently:

$$\frac{1}{\sigma_i(E)}\frac{\mathrm{d}\sigma_i(E,\Omega)}{\mathrm{d}\Omega}\mathrm{d}\Omega$$

is the probability that a photon of energy E will scatter into angular interval $\mathrm{d}\Omega$ around Ω after interaction type i.

Macroscopic scale: linear attenuation coefficient μ

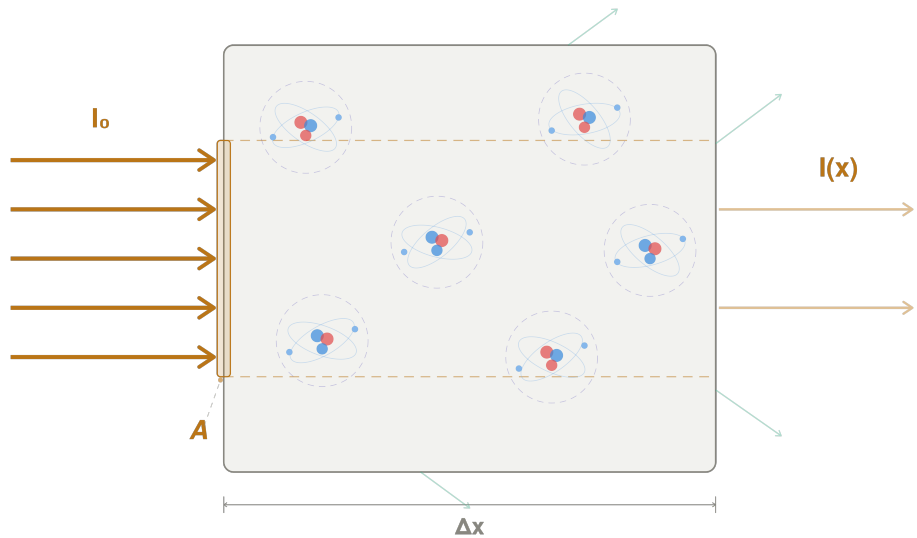
μ is the interaction rate per unit path length governing exponential attenuation, and is determined by the microscopic cross section σ.

If there are n atoms per unit volume, the number of atoms in volume $A \cdot \Delta x$ is

$$n \cdot A \cdot \Delta x$$

and the total *"target area"* blocked by all atoms in that volume is

$$n \cdot A \cdot \Delta x \cdot \sigma$$



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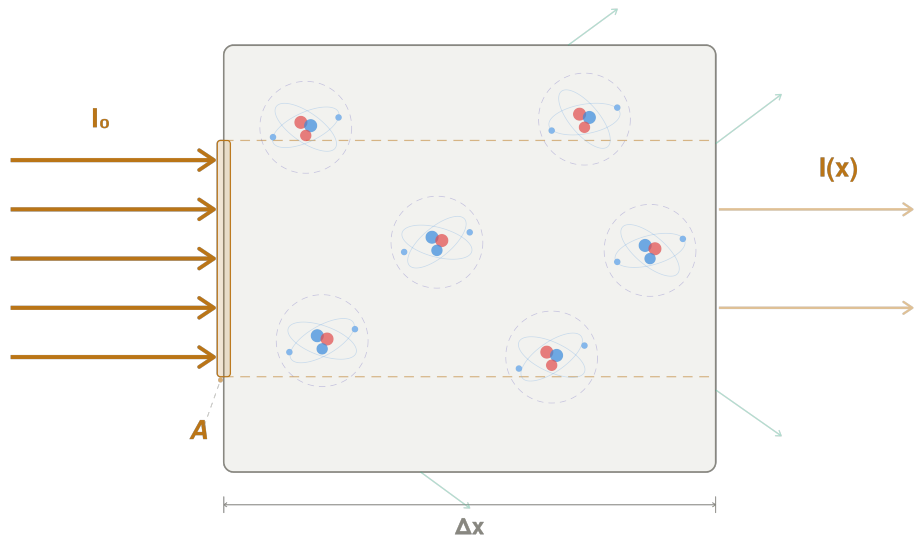
Macroscopic scale: linear attenuation coefficient μ

μ is the interaction rate per unit path length governing exponential attenuation, and is determined by the microscopic cross section σ.

The probability that an incoming particle interacts while crossing Δx is the ratio of total target area to beam area A:

$$P(\Delta x) = \frac{n \cdot A \cdot \Delta x \cdot}{A} = n \cdot \Delta x \cdot \sigma$$



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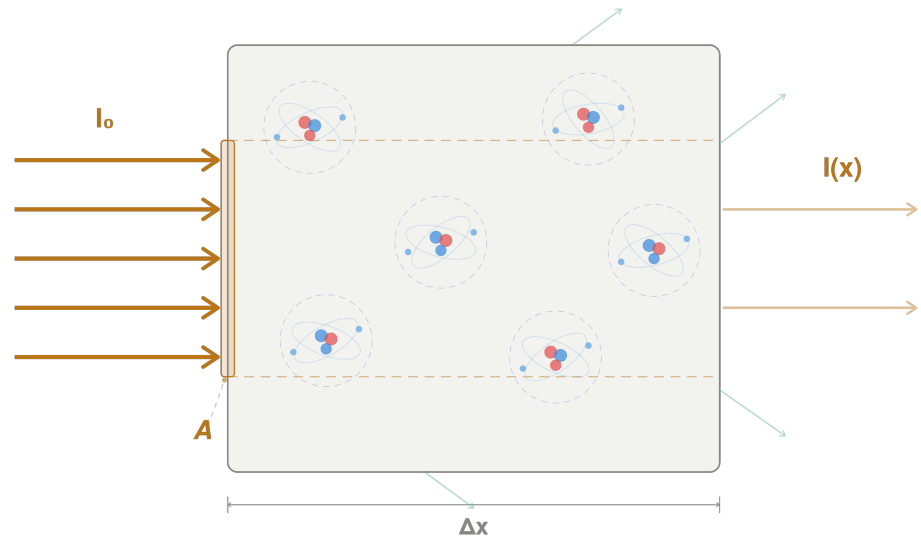
Macroscopic scale: linear attenuation coefficient μ

μ is the interaction rate per unit path length governing exponential attenuation, and is determined by the microscopic cross section σ.

The change in intensity crossing the slab is proportional to how many particles interact.

For a thin slab thickness $\mathrm{d}x$:

$$dI = -I \cdot P(\mathrm{d}x) = -I \cdot n \cdot \sigma \cdot \mathrm{d}x$$



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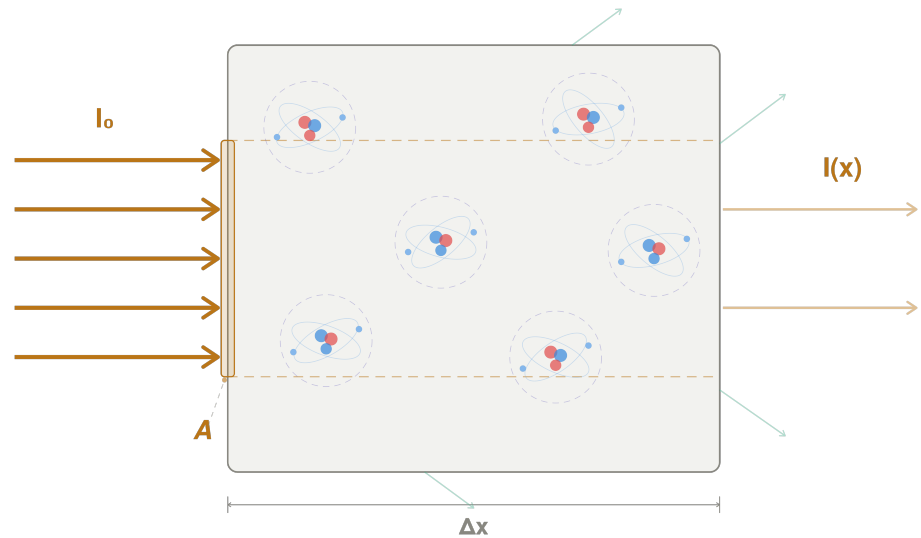
Macroscopic scale: linear attenuation coefficient μ

μ is the interaction rate per unit path length governing exponential attenuation, and is determined by the microscopic cross section σ.

which gives the ODE:

$$\frac{\mathrm{d}I}{\mathrm{d}x} = -I \cdot n \cdot \sigma$$



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Macroscopic scale: linear attenuation coefficient μ

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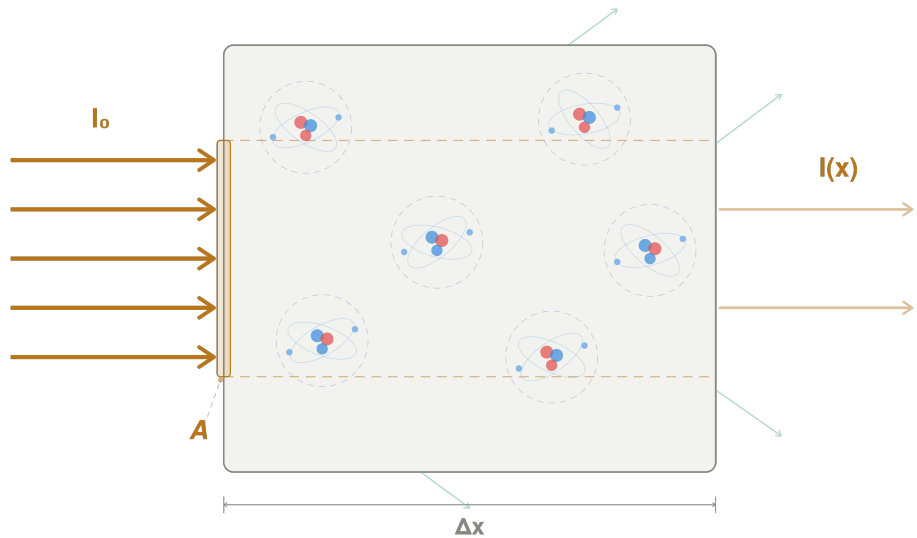
$$\frac{\mathrm{d}I}{\mathrm{d}x} = -I \cdot n \cdot \sigma = -I \cdot \mu$$



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Typical unit: cm^{-1}

Macroscopic scale: linear attenuation coefficient μ

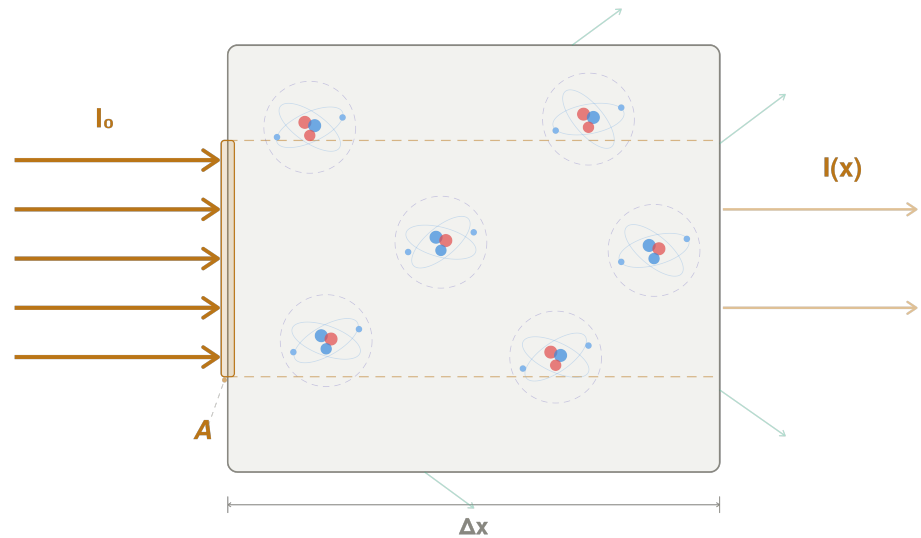
μ is the interaction rate per unit path length governing exponential attenuation, and is determined by the microscopic cross section σ.

which gives the ODE:

$$\frac{\mathrm{d}I}{\mathrm{d}x} = -I \cdot n \cdot \sigma = -I \cdot \mu$$

and integrating from 0 to x with $I(0) = I_\mathrm{o}$ yields the *exponential attenuation law*:

$$I(x) = I_\mathrm{o} \cdot \mathrm{e}^{-\mu x}$$



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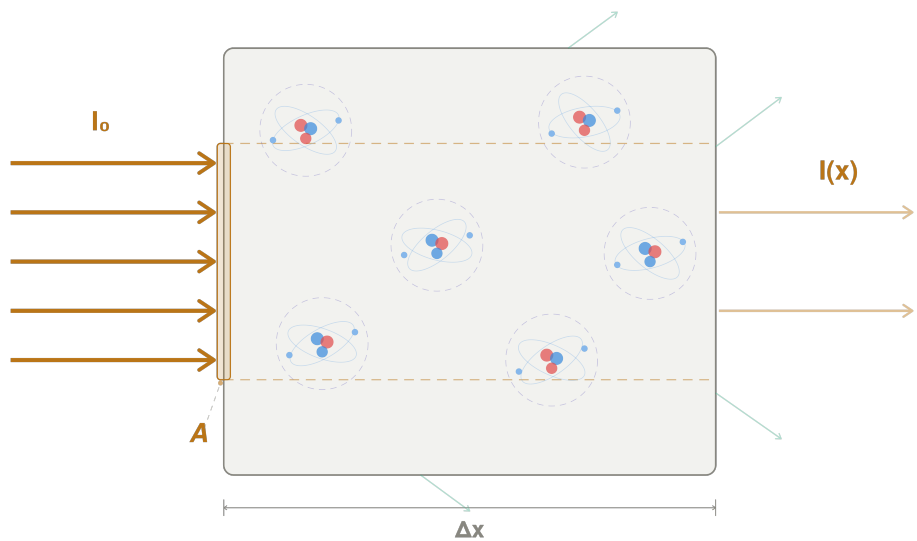
Typical unit: cm^{-1}

Macroscopic scale: mass attenuation coefficient $\frac{\mu}{\rho}$

Removes the dependence on the physical density of the material, making it a more fundamental quantity that depends only on the atomic composition.

- A given material can exist at different densities
- Store the physics once and recover the linear coefficient for any particular density simply by multiplying by ρ
- Varies relatively smoothly across materials, more useful for interpolation and for building physical intuition about how different materials compare



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Typical unit: cm^2/g

Photon transport

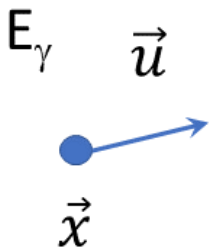
- Photons are neutral particles that travel through matter in straight lines
- Significantly simpler than transport of charged particles, which follow complicated trajectories as they continuously slow down in a medium due to atomic excitation or ionization
- Neutral particles are characterized by their mean-free-path λ while charged particles are characterized by their mass stopping power L/ρ and their range R_{CSDA}
- Since distance to interaction d follows the **exponential distribution**

$$f(d) = \mu\, e^{-\mu\, d}$$

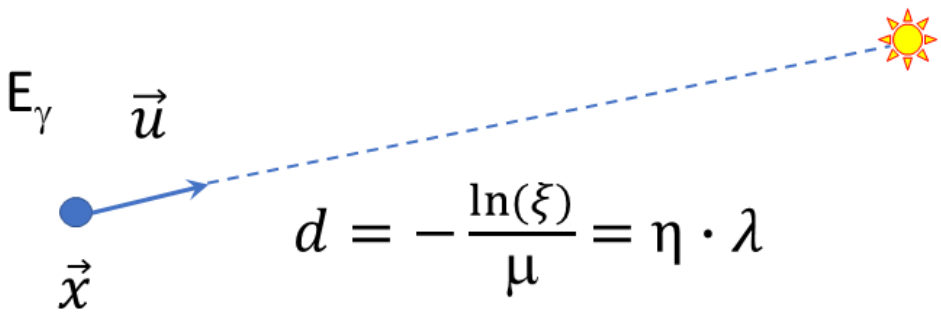
where μ is the **linear attenuation coefficient**, the mean-free-path λ is given by

$$\lambda = \frac{1}{\mu}$$

Photon transport mechanics



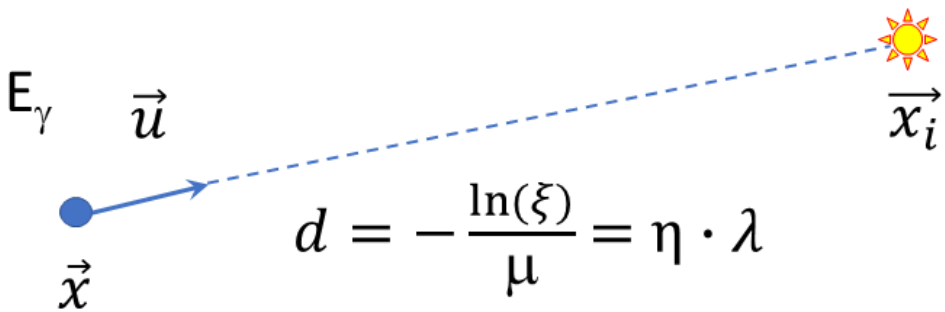
Photon transport mechanics



$$d = -\frac{\ln(\xi)}{\mu} = \eta \cdot \lambda$$

Where $\xi \sim U[0,1)$

Photon transport mechanics

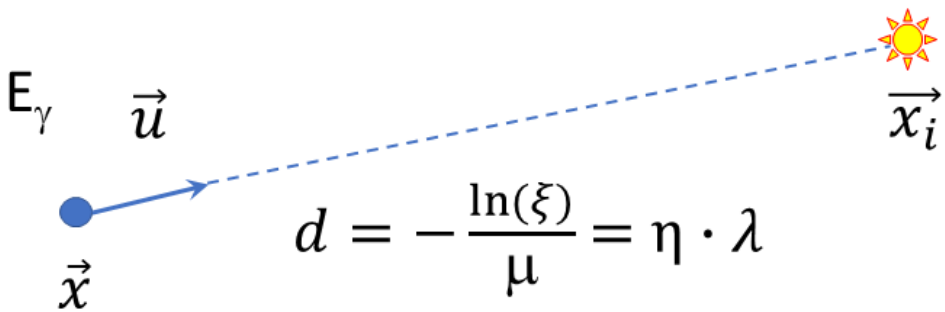


$$\omega_i = \frac{\sigma_i}{\sigma}$$

$$d = -\frac{\ln(\xi)}{\mu} = \eta \cdot \lambda$$

Where $\xi \sim U[0,1)$

Photon transport mechanics



$$d = -\frac{\ln(\xi)}{\mu} = \eta \cdot \lambda$$

Where $\xi \sim U[0,1)$

$$\omega_i = \frac{\sigma_i}{\sigma}$$

<u>Resulting particles</u>

$E_1, \vec{u}_1$

$E_2, \vec{u}_2$

.

.

.

$E_n, \vec{u}_n$

Photon interactions

In the energy range, relevant to Medical Physics, photons interact with surrounding matter via four basic processes:

- **Materialization** into electron–positron pairs in the nuclear and electron field
- **Compton (incoherent)** scattering of atomic electrons
- **Photoelectric** absorption
- **Rayleigh (coherent)** scattering of atoms or molecules

On the high-energy end a fifth interaction possibility arises:

- **Photonuclear** attenuation

Of concern in high-energy clinical linacs (above 10 MV).

General aspects of photon interactions with matter

- Indirectly ionizing through the generation of secondary particles (electrons, positrons, neutrons).
- Involve either the nuclei or orbital electrons of the atoms in the medium.
- Interactions with nuclei might be direct `photon-nucleus` (photo-disintegration), or with the `electrostatic field` of the nucleus (pair production).
- Interaction with `losely-`bound orbital electrons ($E_{\mathrm{B}} \ll h\nu$): Thomson scattering, Compton scattering, tripplet production.
- Interaction with `tightly-`bound orbital electrons ($E_{\mathrm{B}} \lesssim h\nu$): photo-electric effect, Rayleigh scattering, bound Compton scattering.

Photon cross sections in EGSnrc

- **By default** the total cross sections for pair and triplet production, photoabsorption and Rayleigh from the NIST photon cross sections database XCOM, and the total cross section for Compton is derived from a theoretical model.
- User has the option to use **alternative cross sections,** such as:
  - EPDL-97 (Evaluated Photon Data Library 1997) maintained and distributed by Lawrence Livermore National Laboratory (LLNL)
  - Storm & Israel 1972

  To select cross sections, add

  ```
  photon cross sections = xcom | si | epdl | [user_prefix]
  ```

  to the transport parameter section of the input file.
- Users may include **their own tabulations** using `user_prefix` which can be any string, for instance `joe`. These can be requested in the MC transport parameters input block

  ```
  photon cross sections = joe
  ```

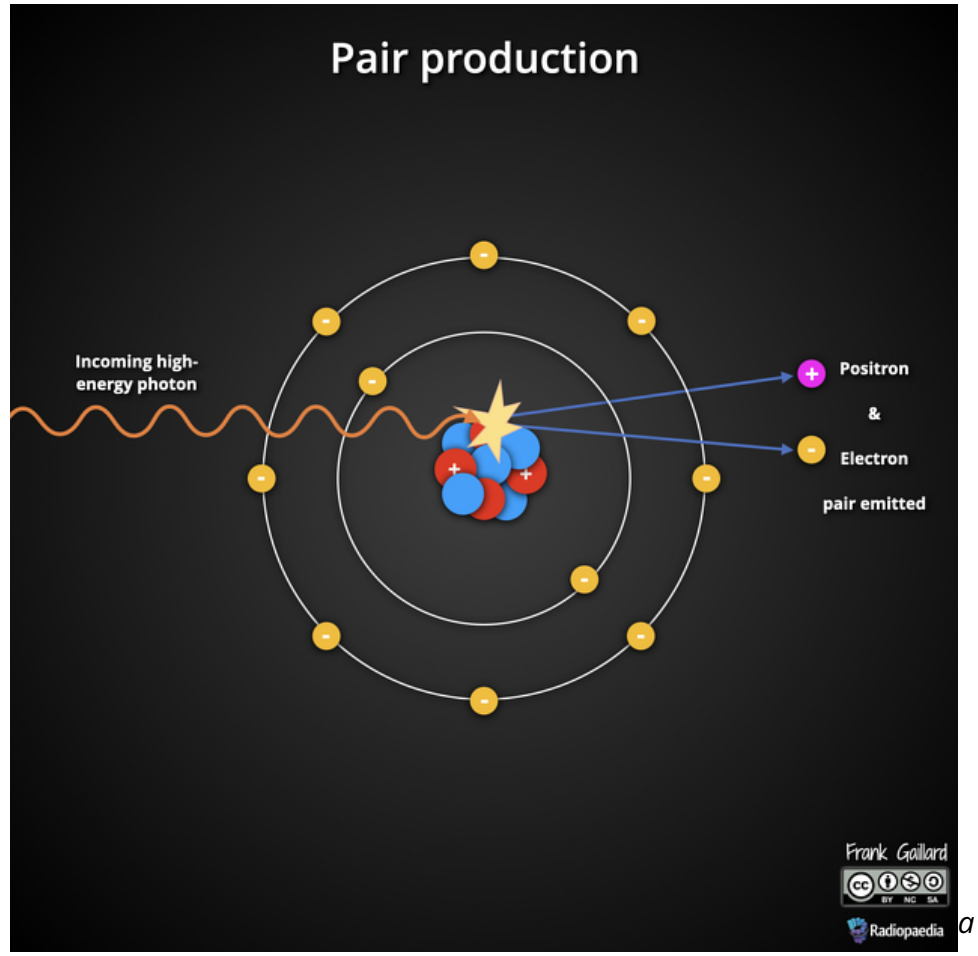
- Cross section files are located in folder `$HEN_HOUSE/data/`

Photon transport parameters

```
:start MC transport parameters:

   # Photon data
   Global Pcut                        =  AP # in MeV
   Photon cross sections              =  xcom
   Pair cross sections                =  BH
   Pair angular sampling              =  Simple
   Triplet production                 =  Off
   Compton cross sections             =  default
   Bound Compton scattering           =  norej
   Radiative Compton corrections      =  Off
   Rayleigh scattering                =  On
   Atomic relaxations                 =  eadl
   Photoelectron angular sampling     =  On
   Photonuclear attenuation           =  Off
   Photonuclear cross sections        =  default

   # Electron transport parameters not included

:stop MC transport parameters:
```

Materialization into e^+e^- pairs: pair production

- Only possible when $k \geq 2\,m_0 = 1.022\ \mathrm{MeV}$
- Cross section scales as Z^2
- Dominates at **high photon energies**



[a]

[a] Case courtesy of Frank Gaillard. Radiopaedia.org, rID: 80801

Pair production is always “On” in EGSnrc

The cross sections can be selected from:

`Pair cross sections = BH | nrc`

`BH (default):` **Bethe-Heitler (BH)** cross sections based on extreme relativistic first Born approximation, Coulomb corrected above 50 MeV.

`nrc:` Option to use an exact formulation based on the exact PWA calculations by Øverbø, Mork and Olsen (OMO) for the unscreened nuclear potential modified by a multiplicative screening correction extended at NRC up to 85 MeV.

Pair production angular distribution

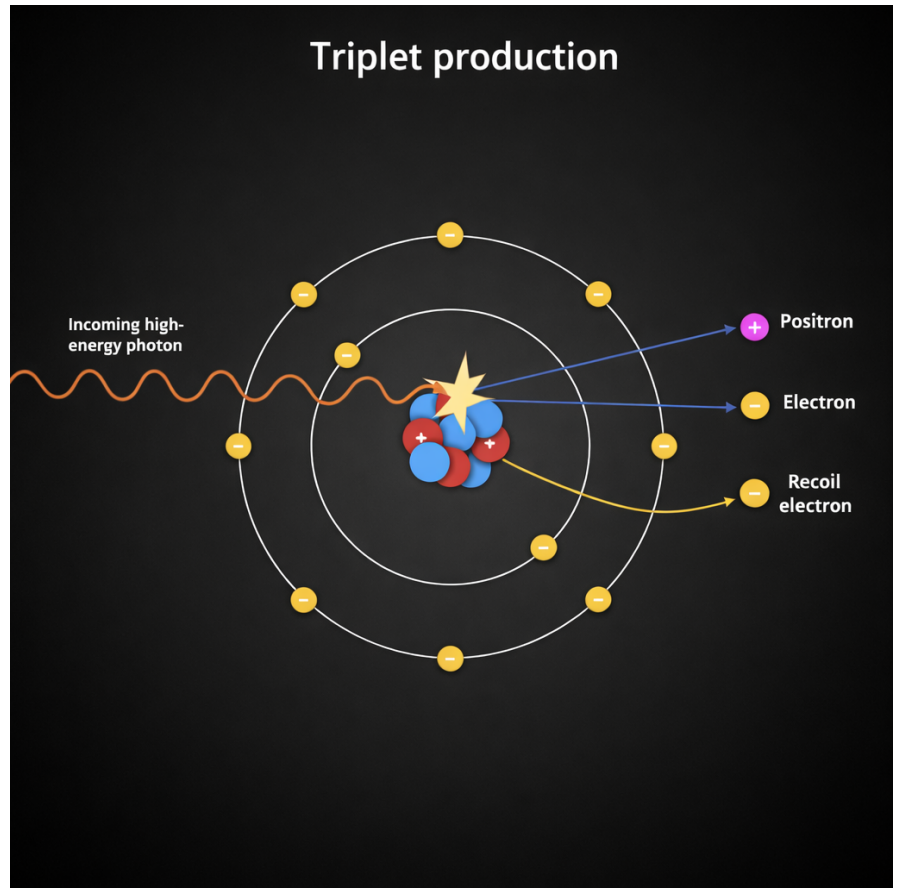
You can select the **angular sampling** method:

```
Pair angular sampling = KM | Simple | Off
```

- `KM`: The more sophisticated distribution from Koch and Motz is derived from an extreme relativistic approximation. It is better at high energies, above 50 MeV.
- `Simple`: The default selection (based on the leading term of the angular distribution) is most likely good enough for any application. The additional CPU cost compared to sampling a fixed angle is negligible.
- `Off`: Using the fixed angle approximation (inherited from EGS4) is never a good idea.

Materialization into e^+e^- pairs: triplet production

- Only for $k \geq 4\,m_0 = 2.044\ \mathrm{MeV}$
- Cross section scales as Z
- **Always** much weaker than pair production (except for low Z)
- "Off" in EGSnrc by default
- Explicit triplet production has a negligible impact in all cases studied

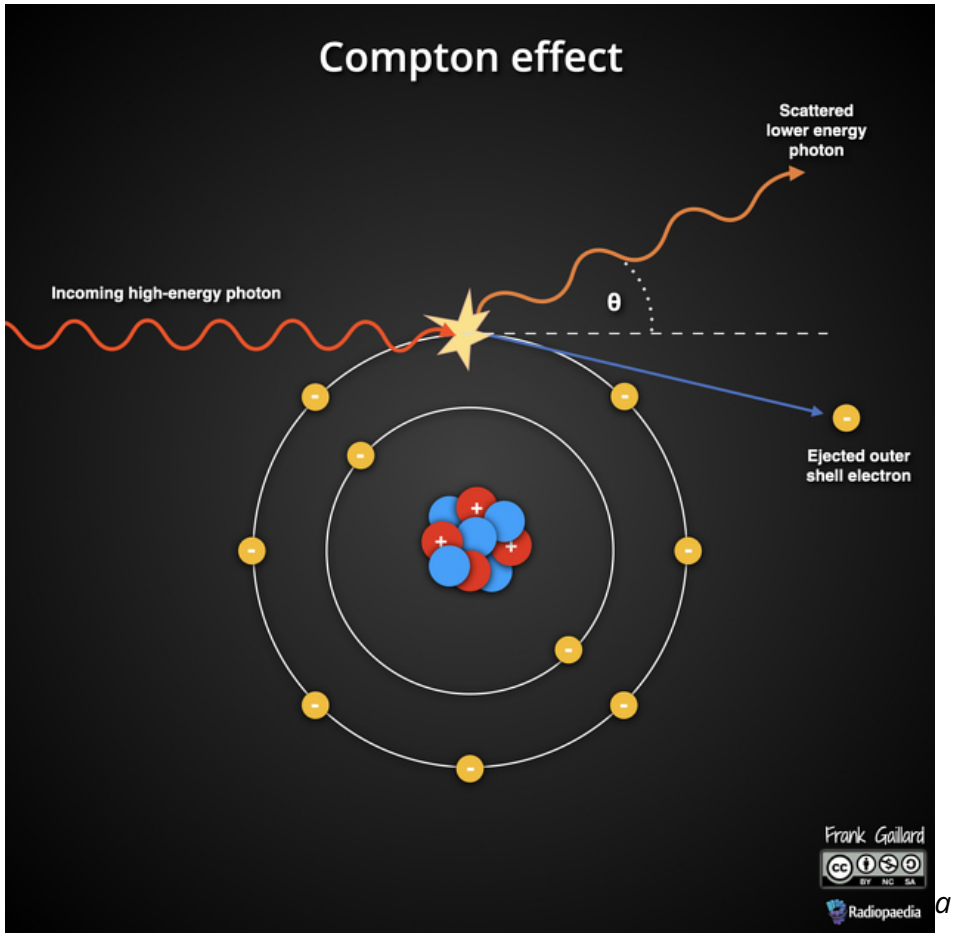


[a] [b]

[a] Illustration generated using AI
[b] Modified from Frank Gaillard. Radiopaedia.org, rID: 80801

Compton (incoherent) scattering

- Occurs for **all** photon energies
- Cross section scales as Z
- Dominates at **intermediate photon energies**
- Electron binding reduces the Compton cross-section
- Binding effects are negligible for energies above a few hundred keV.



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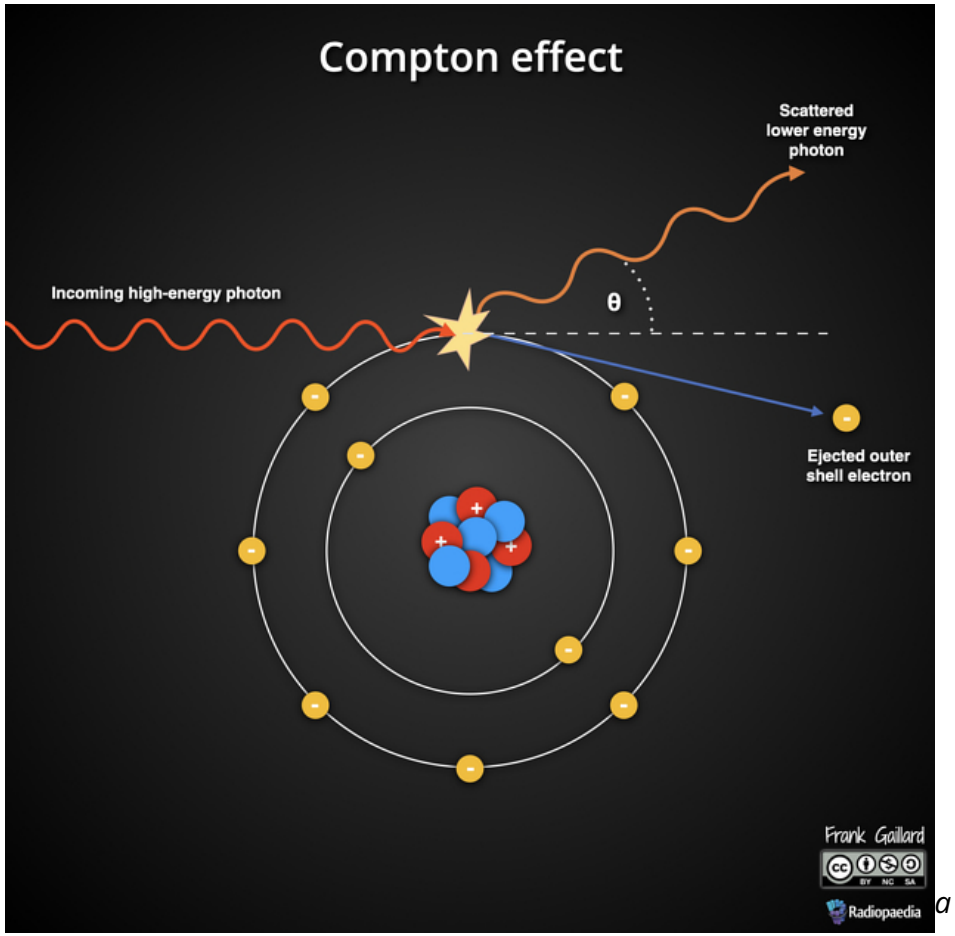
[a] Case courtesy of Frank Gaillard. Radiopaedia.org, rID: 80801

Compton (incoherent) scattering

- By default EGSnrc uses the **relativistic impulse approximation (RIA),**
- **RIA** takes into account binding and Doppler broadening.
- **RIA** is about 2 times slower than sampling interactions with free electrons.
- Significant for photon-only simulations.



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[a] Case courtesy of Frank Gaillard. Radiopaedia.org, rID: 80801

Bound Compton scattering in EGSnrc

Optionally, one can select a different incoherent scattering model:

`Bound Compton scattering = off | simple | on | norej`

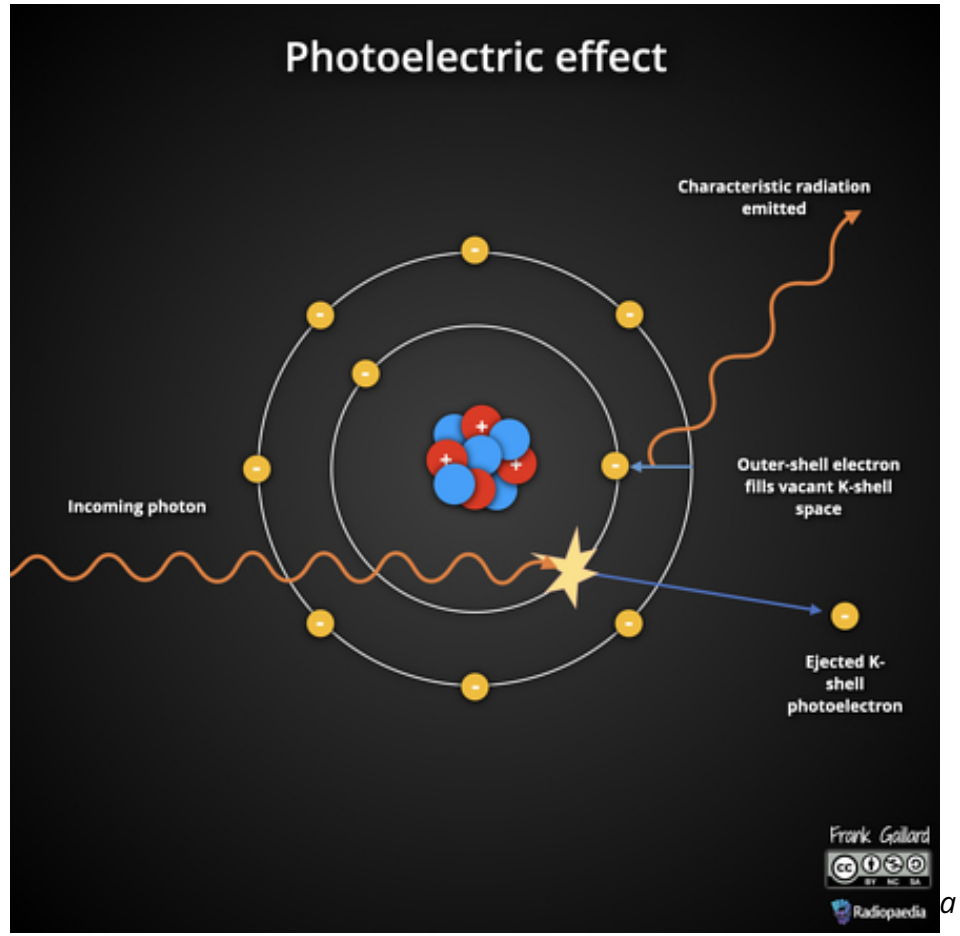
- `off`: Klein-Nishina sampling (free electron).
- `simple`: simple bound model, no Doppler broadening (similar to XCOM).
- `on`: relativistic impulse approximation.
- `norej`: same as `on`, but more efficient (`default`).

To use alternative Compton cross sections, e.g., from the file `joe_compton.data` , add `Compton cross sections = joe` to the transport parameter section of the input file.

EGSnrc allows the user to include radiative corrections for Compton scattering in the one-loop approximation. These corrections are by default turned off.

Photoelectric absorption

- Occurs at all energies, but probability is exceedingly small at high energies
- Cross section scales as Z^n, $n \sim 3 \cdots 5$
- Dominates at **low photon energies**



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[a] Case courtesy of Frank Gaillard. Radiopaedia.org, rID: 80801

Photoelectron direction

By default sampled from the Sauter distribution*:

- Valid only for the K-shell
- Derived from an extreme relativistic approximation ($\beta \approx 1$)
- Good agreement with absorption measurements down to 92 keV

The photoelectric effect is only dominant at low energies, where the angular distribution of electrons rapidly becomes isotropic due elastic scattering. But the CPU cost is negligible, so angular sampling of the photoelectron is On by default.

turned On or Off using:

```
Photoelectron angular sampling = On | Off
```

When turned off, photoelectron inherits the photon's direction

* See EGSnrc manual for further details

Atomic relaxations

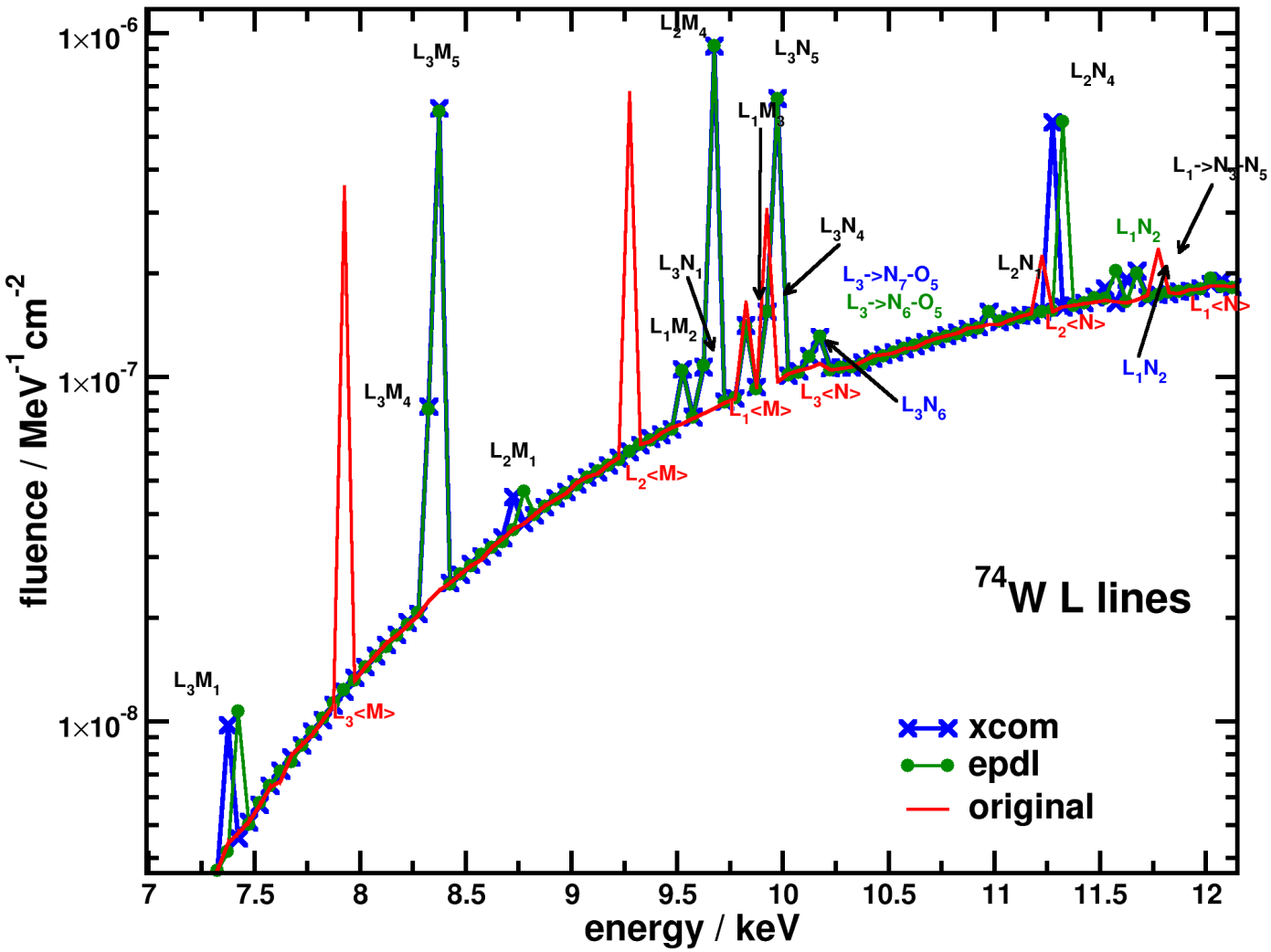
The input options for atomic relaxations are:

`Atomic relaxations = Off | [On | eadl] | simple`

EADL $\equiv$ Evaluated Atomic Data Library by LLNL

- Relaxation cascade of inner shell vacancies generates fluorescent photons, Auger or Coster-Kronig electrons. Of importance if their ranges are not small compared to the geometrical scale of the problem.
  - Currently in photo-absorption, bound Compton scattering, electron impact ionization
  - Future: triplet production, electron-electron bremsstrahlung
- If `simple`: All radiative and non-radiative transitions to/from K-, LI-, LII-, LIII-, and "average" M and N-shells with binding energies above 1 keV.
- If `On|eadl`: All radiative and non-radiative transitions to/from all available shells with binding energies above 1 keV.
- Cannot be `Off` when bound Compton or electron impact ionization requested!

Atomic relaxations



Photoelectric interaction with atomic shells

By default, the interacting shell is explicitly sampled using energy independent interaction probabilities ν_i with the K-, LI-, LII-, LIII-, and "average" M-shell with binding energies above 1 keV.

The input entries in the `Transport Parameters` input block influencing how the interaction occurs are:

```
Photon cross sections = xcom | epdl | si
Atomic relaxations    = simple | eadl
```

Atomic shells involved in the interaction depend on the atomic relaxation option selected:

Relaxation	simple	EADL
Initial vacancy	K,L1..L3,<M>	K,L1..L3
Final vacancy	K,L1..L3,<M>,<N>	K,L1..L3,M1..M5,N1..N7..

Photoelectric interaction with atomic shells

Option to use Sabbatucci and Salvat **renormalized** * shell-wise cross sections allowing to determine energy-dependent interaction probabilities for all available shells by setting:

```
Photon cross sections = mcdf-xcom | mcdf-edpl
Atomic relaxations    = eadl
```

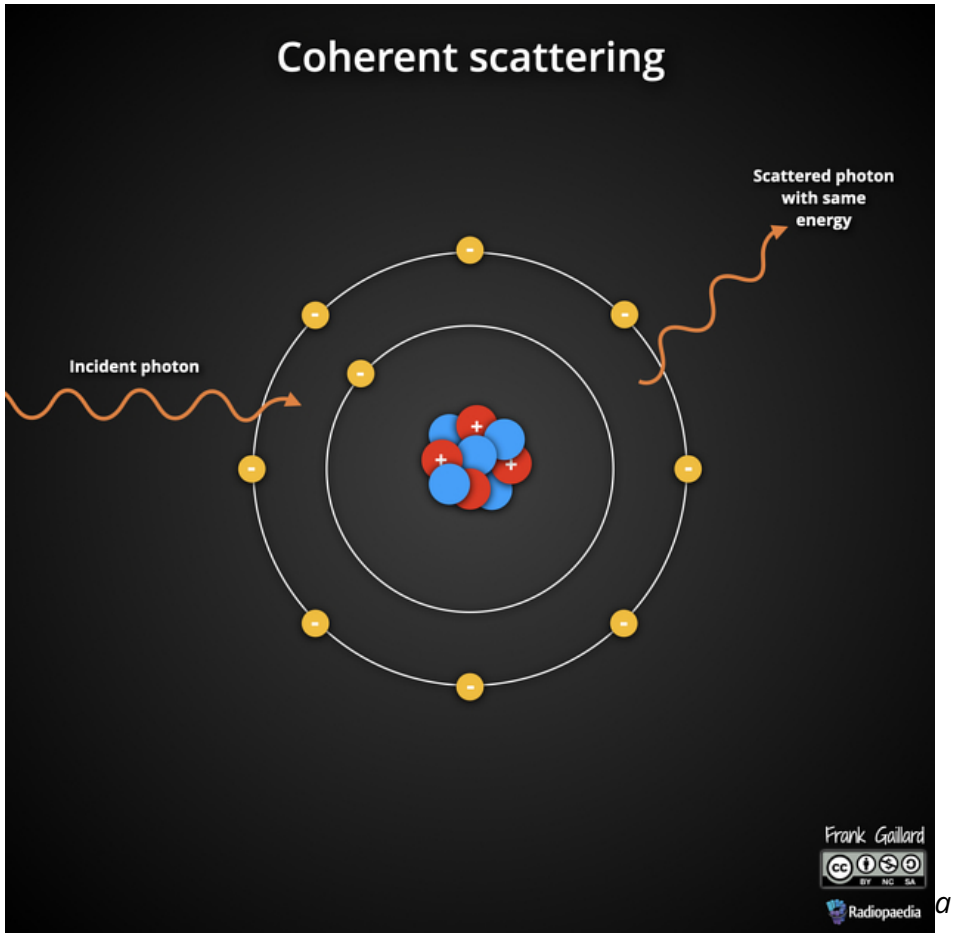
Atomic shells involved in the interaction:

Photoeffect	MCDF
Initial vacancy	K,L1..L3,M1..M5,N1..N7
Final vacancy	K,L1..L3,M1..M5,N1..N7..

* In principle, one can improve independent-electron cross sections using an energy-independent factor, obtained from atomic-structure calculations such as multi-configuration Dirac-Fock (MCDF) self-consistent model.

Rayleigh (coherent) scattering

- Occurs for **all** photon energies
- But probability is extremely small at high energies
- Cross sections scale as Z^2 (small angles)
- Cross sections scale as Z^3 (large angles)
- **Never** dominates



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[a] Case courtesy of Frank Gaillard. Radiopaedia.org, rID: 80801

Rayleigh scattering is “On” in EGSnrc by default

This effect can be turned on and off:

```
Rayleigh scattering = Off | On | custom
```

EGSnrc uses **atomic form factors** from the tabulations by Hubbell and Øverbø, which are derived from atomic form factors using the independent atom approximation.

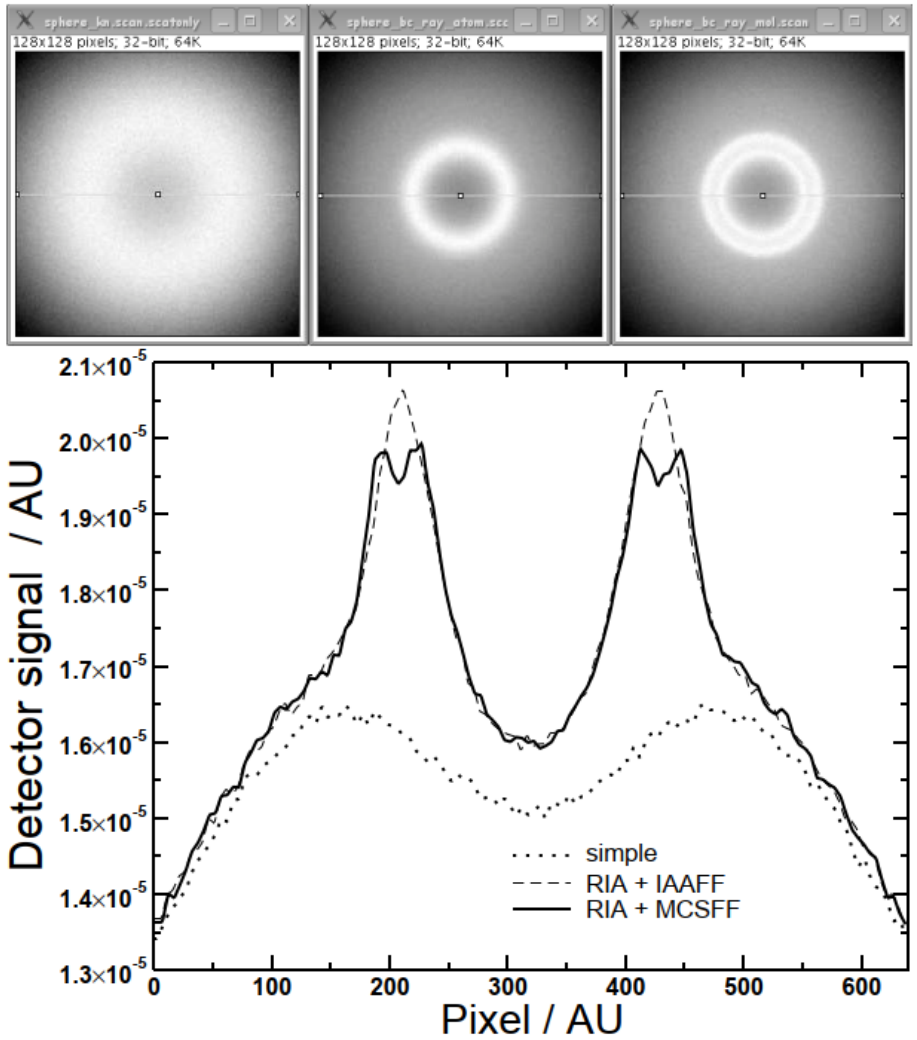
The `custom` option allows users to employ **their own form factor tabulations.** For example, to use the form factor data found in `file1` for the medium `medium1`, etc.:

```
Rayleigh scattering = custom
ff media names = medium1 medium2 ...
ff file names  = file1 file2 ...
```

Measured molecular form factors for 15 substances (including water) from Peplow and Verghese are distributed with EGSnrc.

Only use Rayleigh scattering when using bound Compton scattering: there is no such thing as a photon elastically scattering off a free electron!

Molecular form factors in imaging



Photonuclear attenuation

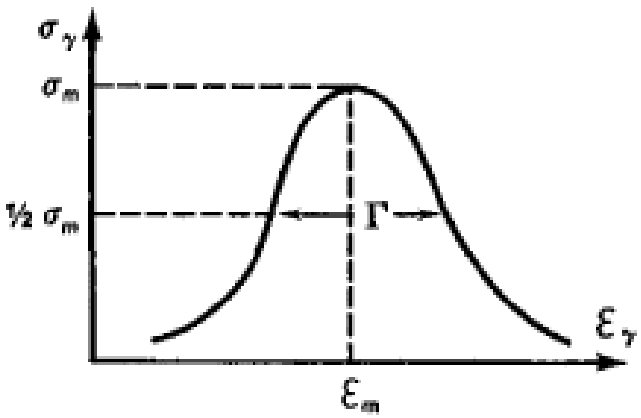
It is possible to account for photon attenuation due to photonuclear interactions (E. Ali):

- IAEA photonuclear cross sections by E. Ali (`iaea_photonuc.data`). Although user-provided cross sections can be used.
- The gamma mean free path (GMFP) in the medium is shortened accordingly.
- Photon is discarded without energy deposition and without generating secondary particles.

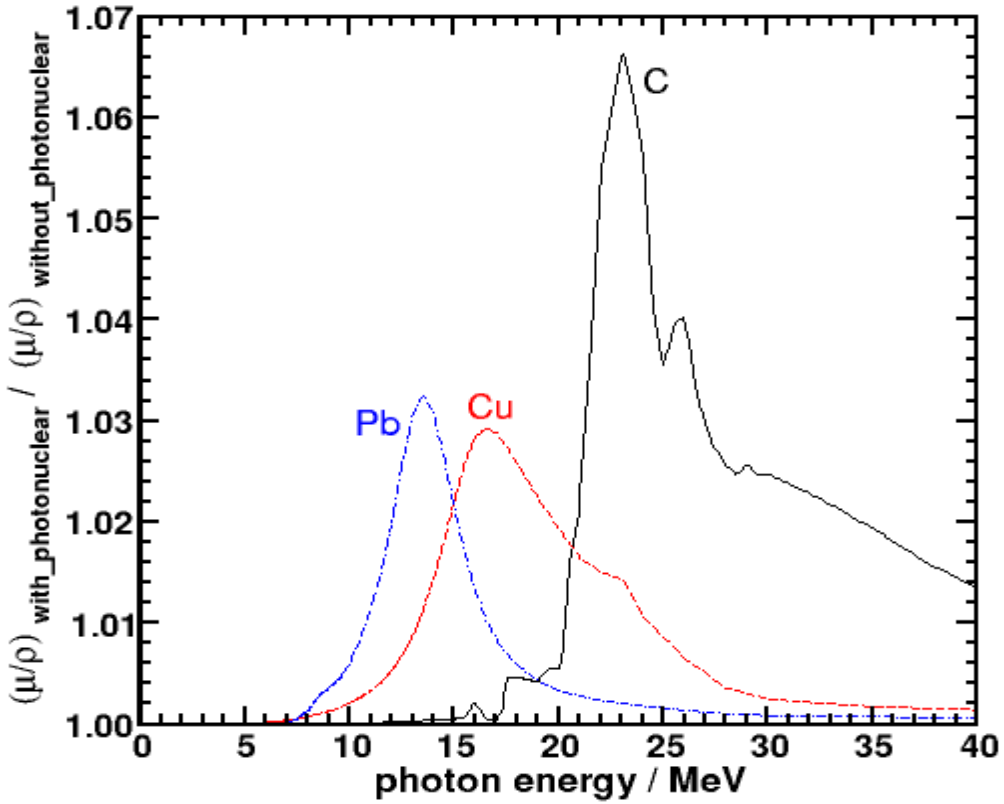
The user's inputs for the photonuclear option are handled in a way similar to the inputs for typical EGSnrc transport parameters:

```
Photonuclear attenuation = OFF (default) | ON | [ON|OFF] in regions
start region = ir1, ir3, . . .
stop region  = ir2, ir4, . . .
Photonuclear cross sections= photonuc_xsections
```

Photonuclear attenuation



The photonuclear cross sections have a broad maximum known as the giant resonance usually spaning about 4 MeV to 8 MeV.



This interaction is of significance for photon energies of the order of tens of MeV.

Default photon transport parameters

```
:start MC transport parameters:

   Photon cross sections            =  xcom # mcdf-xcom?
   Compton cross sections           =  default
   Global Pcut                      =  AP # in MeV
   Pair cross sections              =  BH
   Pair angular sampling            =  Simple
   Triplet production               =  Off
   Bound Compton scattering         =  norej
   Radiative Compton corrections    =  Off
   Rayleigh scattering              =  On
   Atomic relaxations               =  eadl
   Photoelectron angular sampling   =  On
   Photonuclear attenuation         =  Off
   Photonuclear cross sections      =  default

:stop MC transport parameters:
```

Optimum parameters for clinical MV photon beams

```
:start MC transport parameters:

   Photon cross sections              =  xcom
   Compton cross sections             =  default
   Global Pcut                        =  0.010 # in MeV
   Pair cross sections                =  BH
   Pair angular sampling              =  Simple
   Triplet production                 =  Off
   Bound Compton scattering           =  Off
   Radiative Compton corrections      =  Off
   Rayleigh scattering                =  Off
   Atomic relaxations                 =  Off
   Photoelectron angular sampling     =  Off
   Photonuclear attenuation           =  Off
   Photonuclear cross sections        =  default

:stop MC transport parameters:
```

Saves about 25 % CPU time for a 6 MV and a 16 MV beam.

Media definition

Materials can be defined:

- in a general material file
  - See for instance `$HEN_HOUSE/pegs4/data/material.dat`
- directly in the input file

```
:start media definition:
  AE=0.521
  UE=50.511
  AP=0.01Each medium defined here
  UP=5

  # Define media here

:stop media definition:
```

- PEGS4-generated file (legacy)
  - Running stand-alone tool PEGS4

Defining materials directly in the input file

```
:start media definition:
  AE=0.521
  UE=50.511
  AP=0.01
  UP=5
  :start graphite:
    elements = C
    number of atoms = 1
    bulk density = 1.70 # g/cm^3 used for transport
    local density = 2.656 # g/cm^3 used for density corrections
    ivalue = 0.000081 # MeV, optional
    output density file = carbon.density # full path to file, optional
  :stop graphite:
  :start water:
    elements = H, O
    number of atoms = 2, 1
    bulk density = 0.998 # g/cm^3 used for transport
    ivalue = 0.000078 # MeV, optional
  :stop water:
  :start air:
    mixture compounds = C, N, O, Ar
    mass fractions = 0.00012400012, 0.75526776, 0.23178123, 0.012827012
    rho = 0.00120479 # g/cm^3 both local and bulk set to this value
  :stop air:
:stop media definition:
```

Medium definition via density correction file

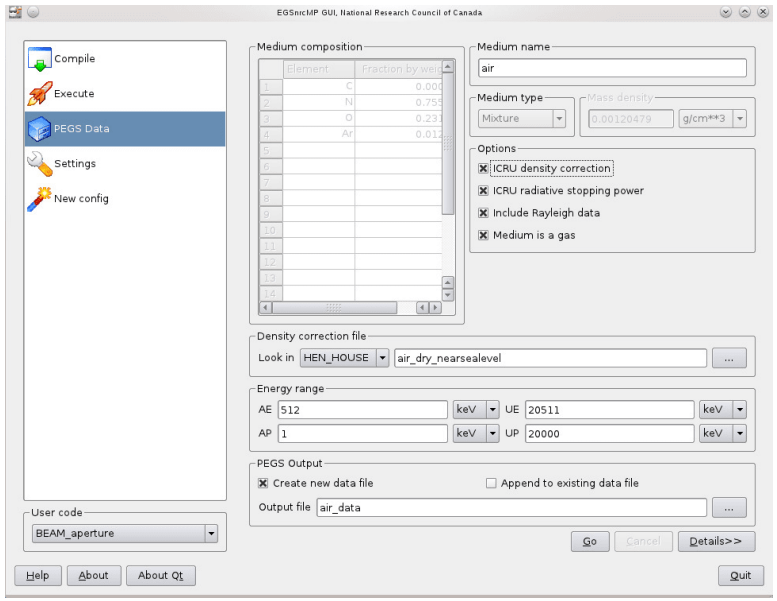
EGSnrc provides *density-effect* corrections to account for the polarization of the medium by the passing charged particle.

```
:start media definition:
    AE=0.521
    UE=50.511
    AP=0.01
    UP=50.

    :start H2O521ICRU:
        density correction file=water_liquid
    :stop H2O521ICRU:

    :start AIR521ICRU:
        density correction file= air_dry_nearsealevel
    :stop AIR521ICRU:
    ...
    :start another:
      ...
    :stop another:
:stop media definition:
```

Location: `$HEN_HOUSE/pegs4/density_corrections/`

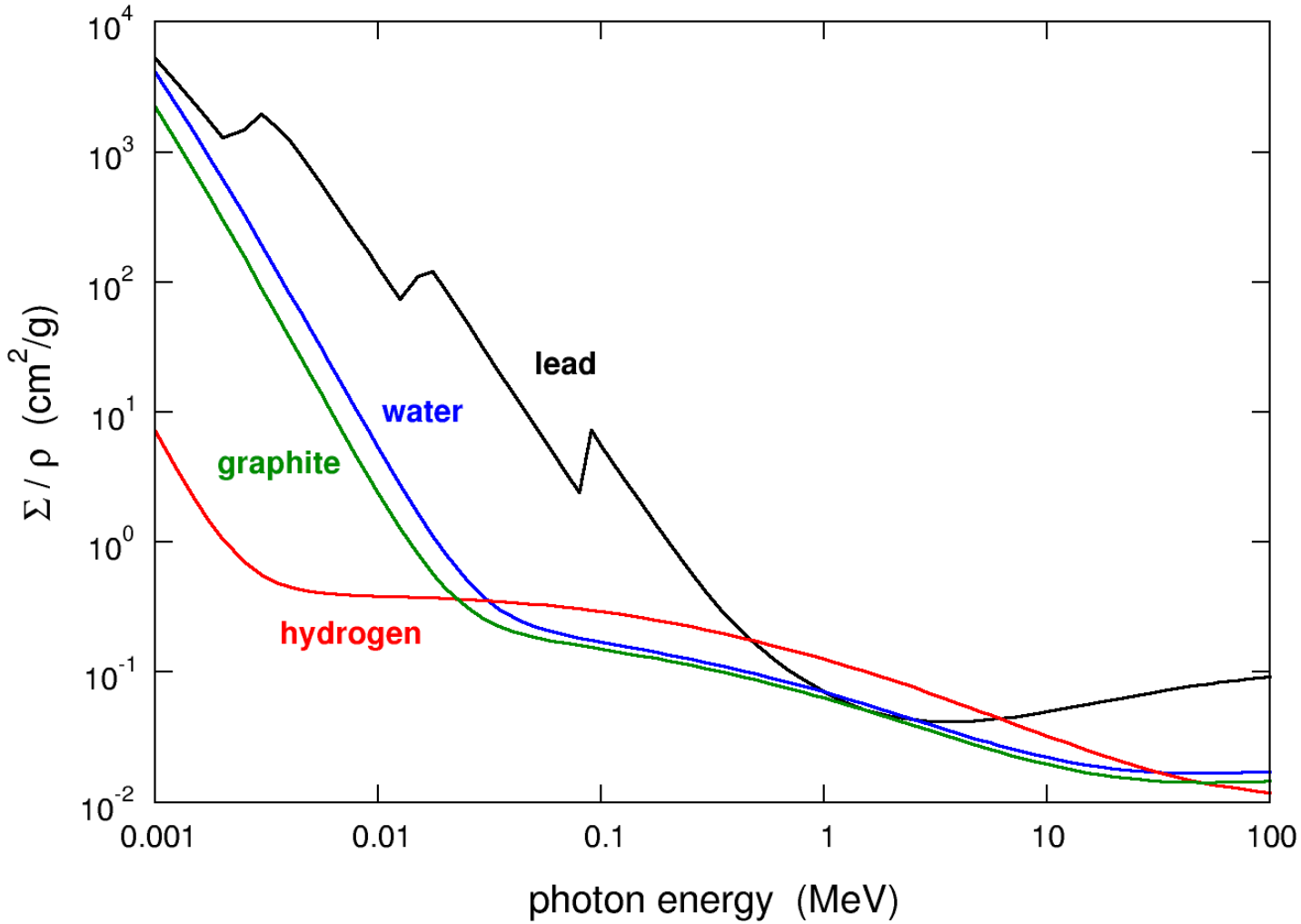
PEGS4: Data Preprocessor for EGS4

- Stand-alone MORTRAN code developed to generate simulation data for EGS4 due to efficiency reasons
  (`It was the 70s and 80s!!!`)
- User can still use PEGS4 data sets.
- There is an app for that: **egs_gui**



Total photon cross sections

(EGSnrc application `examin`)



Take-home messages

- EGSnrc uses `state-of-the-art` photon physics models
- Options to use different models allows comparison with other MC codes and historical results
- Options to use simpler models increasing calculation speed
- Users have access to the EGSnrc data: Useful to verify your simulation setup, build physical intuition about which interactions dominate at a given energy, and troubleshoot unexpected results.

ICTP-SAIFR EGSnrc Monte Carlo School for Medical Physics

IFT-UNESP, São Paulo, Brazil

March 30 – April 3, 2026

LECTURE 05 · EGSnrc course

Photon physics

Ernesto Mainegra-Hing · National Research Council Canada, Metrology Research Centre